Multivariate Calculus Lecture Notes
Section 14.2 – Limits and Continuity in Higher Dimensions

I. Continuity of multivariable functions
   a. A function of two variables is *continuous* at a point (a, b) if $\lim_{(x,y)\to(a,b)} f(x, y) = f(a, b)$.
   b. But what does it mean to take a limit of a multivariable function?

II. Limits of multivariable functions
   a. How do you evaluate $\lim_{(x,y)\to(0,0)} \dfrac{x^2 - y^2}{x^2 + y^2}$?
      i. The function value doesn't exist at $(0, 0)$, so that doesn't help.
      ii. Show graph on Mathematica.
      iii. It looks like there's a discontinuity at $(0, 0)$.
         1. Along the x-axis it looks like the limit is 1.
         2. Along the y-axis it looks like the limit is -1.
         3. How do you prove this?
      iv. Show contour plot on Mathematica.
         - Since multiple contour lines appear to converge on the origin it looks like we have a discontinuity there.
      v. Finding limits along different paths
         1. To find the limits along the x-axis, fix y at 0 and let x approach 0.
         2. Thus, along the x-axis, we get $f(x, y) \to x^2/x^2 = 1$.
         3. And, along the y-axis, we get $f(x, y) \to -y^2/y^2 = -1$.
         4. Since the limits from two different directions are different the limit does not exist.
   b. Here's the idea: In order for $\lim_{(x,y)\to(a,b)} f(x, y)$ to be some value L the function values have to approach L as (x, y) approaches (a, b) from any direction.
      i. More formally, $\lim_{(x,y)\to(a,b)} f(x, y) = L$ if $f(x, y)$ is as close to L as we want whenever (x, y) is sufficiently close to (a, b).
      ii. We measure the distance from $f(x, y)$ to L by $|f(x, y) - L|$.
      iii. We measure the distance from (x, y) to (a, b) by $\sqrt{(x - a)^2 + (y - b)^2}$.
      iv. The formal definition of the limit of a multivariable function is the following: $\lim_{(x,y)\to(a,b)} f(x, y) = L$ if, for every $\epsilon > 0$ there exists $\delta > 0$ such that if $0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta$ then $|f(x, y) - L| < \epsilon$.
      v. In the previous example:
         1. The number 1 can't be the limit, from the definition, because if you give me $\epsilon = 0.1$ then I can't squeeze the disk of radius δ tight enough around (a, b) to force all the function values to be within 0.1 units of 1.
         2. More generally, the limit can't exist because there are points arbitrarily close to $(0, 0)$ that have a function value of -1 and points arbitrarily close that have a function value of 1.
   c. More examples
      i. Find, if it exists, $\lim_{(x,y)\to(0,0)} \dfrac{xy}{x^2 + y^2}$.
         1. The value $f(0, 0)$ does not exist.
         2. Show graph on Mathematica.

3. Approach along x-axis yields $f(x,y) \to 0/x^2 = 0$.
4. Approach along y-axis yields $f(x,y) \to 0/y^2 = 0$.
5. But this doesn't necessarily mean that the limit is 0; in fact, it looks like a diagonal approach will produce a different limit.
6. Approach along the line $y = x$ yields $f(x,y) \to x^2/(2x^2) = 1/2$.
7. Thus the limit does not exist.

ii. Find, if it exists, $\lim_{(x,y)\to(0,0)} \dfrac{xy^2}{x^2+y^4}$.

1. The value of $f(0,0)$ does not exist.
2. Show graph on Mathematica.
3. Approach along any line through the origin $y = mx$:

$$f(x,y) \to \frac{m^2x^3}{x^2+m^4x^4} = \frac{m^2x}{1+m^4x^2} = 0.$$

4. It looks like approaching along the parabola $x = y^2$ will yield something different, though.
5. If we do this we get $f(x,y) \to \dfrac{y^4}{y^4+y^4} = \dfrac{y^4}{2y^4} = \dfrac{1}{2}$.
6. Thus the limit does not exist.

iii. Find, if it exists, $\lim_{(x,y)\to(0,0)} \dfrac{3x^2y}{x^2+y^2}$.

1. Once again, $f(0,0)$ does not exist.
2. Show graph on Mathematica.
3. Show contour plot on Mathematica.
4. It looks like the limit is actually 0, so let's try to prove this.
5. Say, for example, that we want the function values to be within 0.01 units of 0.
   - So we want to find a δ such that if $0 < \sqrt{x^2+y^2} < \delta$ then $|f(x) - L| < 0.01$.
   - We have

$$|f(x) - L| \le \left|\frac{3x^2y}{x^2+y^2} - 0\right| = \frac{3x^2|y|}{x^2+y^2} \le 3|y| \le 3\sqrt{x^2+y^2}.$$

   - We're guaranteed, by hypothesis, that the final quantity is less than 3δ.
   - So, if we pick δ to be $0.01/3$ then we will have

$$|f(x) - L| \le \left|\frac{3x^2y}{x^2+y^2} - 0\right| = \frac{3x^2|y|}{x^2+y^2} \le 3|y| \le 3\sqrt{x^2+y^2} < 3\delta = 3\left(\frac{0.01}{3}\right) = 0.01.$$

   - Thus, if we force the (x,y) values to be within $0.01/3$ units of $(0,0)$ then the function values are guaranteed to be within 0.01 units of 0.
6. More generally, how could we redo this argument for arbitrary ϵ?
   - The entire argument is the same except that at the end we would need to replace 0.01 with ϵ.
   - So we pick $\delta = \epsilon/3$ and repeat the rest of the argument.
   - This proves that $\lim_{(x,y)\to(0,0)} \dfrac{3x^2y}{x^2+y^2} = 0$.

III. Back to continuity of multivariable functions

a. Continuous function properties:

i. If k is real and f, g are continuous at (a,b), then

1. kf is continuous
2. $f \pm g$ is continuous
3. fg is continuous
4. f/g is continuous (unless $g(a,b) = 0$)

ii. If h is continuous at (a,b) and g is continuous at $h(a,b)$ then $g(h(x,y))$ is continuous at (a,b).

1. Mathematically, $\lim_{(x,y)\to(a,b)} g(h(x,y)) = g(h(a,b))$.
2. Composition of continuous functions is continuous.

b. This implies that polynomial and rational functions (where they exist) are continuous.

c. Examples

i. Where is the function $f(x,y) = \dfrac{3x^2y}{x^2+y^2}$ continuous? Everywhere except $(0,0)$.

ii. Where is the function $f(x,y) = \ln(x^2+y^2-1)$ continuous?

1. On the set $x^2+y^2-1 > 0 \Rightarrow x^2+y^2 > 1$
2. So $f(x,y)$ is continuous outside the circle of radius 1 centered at the origin.

d. Using continuity to evaluate limits

i. Thanks to the definition of continuity, evaluating multivariable limits is often as easy as substitution.

ii. Example (#33): Evaluate $\lim_{(x,y)\to(1,-3)} e^{x-y}\ln(x-y)$.

1. The only potential problems with continuity are those for which $x-y < 0$, or, equivalently, $y > x$.
2. Since we don't have that for $(1,-3)$, we can just substitute:

$$\lim_{(x,y)\to(1,-3)} e^{x-y}\ln(x-y) = e^{1-(-3)}\ln(1-(-3)) = e^4\ln 4.$$

Homework: Section 14.2 #1, 6, 7, 12, 13, 14, 21, 30, 34, 43.